

1.2 Rank, column space and null space

1.2.1 Subspaces and dimension

A nonempty set $\mathcal{S} \subseteq \mathbb{R}^n$ is a **subspace** if $\mathbf{u}, \mathbf{v} \in \mathcal{S}$ and $a, b \in \mathbb{R}$ imply $a\mathbf{u} + b\mathbf{v} \in \mathcal{S}$. Vectors $\mathbf{v}_1, \dots, \mathbf{v}_k$ are **linearly independent** if $\sum_j c_j \mathbf{v}_j = \mathbf{0}$ only for $c_1 = \dots = c_k = 0$, that is, if the matrix $[\mathbf{v}_1, \dots, \mathbf{v}_k]$ sends only the zero vector to zero. Their **span** is the set of all their linear combinations. A **basis** of $\mathcal{S}$ is an independent set that spans it.

We use the following facts of elementary linear algebra without proof (Halmos 1958, chapter 1). Every subspace of $\mathbb{R}^n$ has a basis. All bases of a subspace have the same number of elements, its **dimension** $\dim \mathcal{S}$. In an r -dimensional subspace, any r independent vectors form a basis, and any independent set can be extended to a basis. If $\mathcal{S} \subseteq \mathcal{T}$ and $\dim \mathcal{S} = \dim \mathcal{T}$, then $\mathcal{S} = \mathcal{T}$. Finally, if $\mathcal{S} \cap \mathcal{T} = \{\mathbf{0}\}$, the union of a basis of $\mathcal{S}$ and a basis of $\mathcal{T}$ is independent, so the sum $\mathcal{S} + \mathcal{T} = \{\mathbf{u} + \mathbf{v} : \mathbf{u} \in \mathcal{S}, \mathbf{v} \in \mathcal{T}\}$ has dimension $\dim \mathcal{S} + \dim \mathcal{T}$. Such a sum is called **direct** and written $\mathcal{S} \oplus \mathcal{T}$.

1.2.2 The four subspaces of a matrix

An $m \times n$ matrix $\mathbf{A}$ carries two subspaces of $\mathbb{R}^m$ and two of $\mathbb{R}^n$:

$$\begin{aligned} \mathcal{C}(\mathbf{A}) &= \{\mathbf{A}\mathbf{b} : \mathbf{b} \in \mathbb{R}^n\} \subseteq \mathbb{R}^m, & \mathcal{N}(\mathbf{A}) &= \{\mathbf{b} \in \mathbb{R}^n : \mathbf{A}\mathbf{b} = \mathbf{0}\} \subseteq \mathbb{R}^n, \\ \mathcal{C}(\mathbf{A}^\top) &\subseteq \mathbb{R}^n, & \mathcal{N}(\mathbf{A}^\top) &\subseteq \mathbb{R}^m. \end{aligned}$$

By Eq. 1.1.2, $\mathcal{C}(\mathbf{A})$ is the span of the columns, the **column space**, and $\mathcal{C}(\mathbf{A}^\top)$ is the span of the rows, the **row space**. $\mathcal{N}(\mathbf{A})$ is the **null space**. The **rank** of $\mathbf{A}$ is $\text{rank}(\mathbf{A}) = \dim \mathcal{C}(\mathbf{A})$, the largest number of independent columns. The number of independent rows, $\dim \mathcal{C}(\mathbf{A}^\top)$, looks like a different quantity. It is not.

Theorem 1.2.1 — Row rank equals column rank

For every matrix $\mathbf{A}$, $\dim \mathcal{C}(\mathbf{A}^\top) = \dim \mathcal{C}(\mathbf{A})$. That is, $\text{rank}(\mathbf{A}^\top) = \text{rank}(\mathbf{A})$. Moreover, if $\text{rank}(\mathbf{A}) = r \geq 1$ there are an $m \times r$ matrix $\mathbf{B}$ and an $r \times n$ matrix $\mathbf{F}$, both of rank r , with $\mathbf{A} = \mathbf{B}\mathbf{F}$.

Proof. Let $r = \text{rank}(\mathbf{A})$ and let the columns of the $m \times r$ matrix $\mathbf{B}$ be a basis of $\mathcal{C}(\mathbf{A})$. Each column $\mathbf{a}_j$ is a combination $\mathbf{B}\mathbf{f}_j$ of these, so $\mathbf{A} = \mathbf{B}\mathbf{F}$ with $\mathbf{F} = [\mathbf{f}_1, \dots, \mathbf{f}_n]$. By the transposed reading of Eq. 1.1.2, every row of $\mathbf{A} = \mathbf{B}\mathbf{F}$ is a combination of the r rows of $\mathbf{F}$, so $\dim \mathcal{C}(\mathbf{A}^\top) \leq r = \dim \mathcal{C}(\mathbf{A})$. Applying this to $\mathbf{A}^\top$ gives the reverse inequality. Finally $\text{rank}(\mathbf{F}) \leq r$ because $\mathbf{F}$ has r rows, and $r = \dim \mathcal{C}(\mathbf{A}^\top) \leq \dim \mathcal{C}(\mathbf{F}^\top) = \text{rank}(\mathbf{F})$, so $\text{rank}(\mathbf{F}) = r$. ■

The factorization $\mathbf{A} = \mathbf{B}\mathbf{F}$ is a **rank factorization**: $\mathbf{B}$ has **full column rank** (its columns are independent) and $\mathbf{F}$ has **full row rank**. It is a convenient tool for proofs, as Proposition 1.9.1 shows.

Theorem 1.2.2 — Rank–nullity

For an $m \times n$ matrix $\mathbf{A}$, $\text{rank}(\mathbf{A}) + \dim \mathcal{N}(\mathbf{A}) = n$.

Proof. Let $\mathbf{v}_1, \dots, \mathbf{v}_k$ be a basis of $\mathcal{N}(\mathbf{A})$ and extend it to a basis $\mathbf{v}_1, \dots, \mathbf{v}_k, \mathbf{w}_1, \dots, \mathbf{w}_{n-k}$ of $\mathbb{R}^n$. Every $\mathbf{A}\mathbf{b}$ is a combination of $\mathbf{A}\mathbf{w}_1, \dots, \mathbf{A}\mathbf{w}_{n-k}$, since $\mathbf{A}\mathbf{v}_i = \mathbf{0}$. These vectors are independent: if $\sum_j c_j \mathbf{A}\mathbf{w}_j = \mathbf{0}$, then $\sum_j c_j \mathbf{w}_j \in \mathcal{N}(\mathbf{A})$ is a combination of the $\mathbf{v}_i$, and independence of the full basis forces every $c_j = 0$. So $\text{rank}(\mathbf{A}) = n - k$. ■

In a linear model with $n \times p$ model matrix $\mathbf{X}$, $\dim \mathcal{N}(\mathbf{X}) = p - \text{rank}(\mathbf{X})$ counts the directions in which the coefficient vector can move without changing the mean $\mathbf{X}\boldsymbol{\beta}$. $\mathbf{X}$ has full column rank iff $\mathcal{N}(\mathbf{X}) = \{\mathbf{0}\}$.

1.2.3 Rank of products and of $\mathbf{A}^\top \mathbf{A}$ **Proposition 1.2.3**

Let $\mathbf{A}$ be $m \times n$ and $\mathbf{B}$ be $n \times q$.

- (a) $\mathcal{C}(\mathbf{AB}) \subseteq \mathcal{C}(\mathbf{A})$ and $\mathcal{N}(\mathbf{B}) \subseteq \mathcal{N}(\mathbf{AB})$, so $\text{rank}(\mathbf{AB}) \leq \min\{\text{rank}(\mathbf{A}), \text{rank}(\mathbf{B})\}$.
- (b) If $\mathbf{B}$ has full row rank, then $\mathcal{C}(\mathbf{AB}) = \mathcal{C}(\mathbf{A})$. If $\mathbf{A}$ has full column rank, then $\mathcal{N}(\mathbf{AB}) = \mathcal{N}(\mathbf{B})$. So $\text{rank}(\mathbf{AB}) = \text{rank}(\mathbf{A})$ in the first case and $\text{rank}(\mathbf{AB}) = \text{rank}(\mathbf{B})$ in the second. In particular, multiplying by a nonsingular matrix does not change rank.
- (c) $\mathcal{N}(\mathbf{A}^\top \mathbf{A}) = \mathcal{N}(\mathbf{A})$ and $\mathcal{C}(\mathbf{A}^\top \mathbf{A}) = \mathcal{C}(\mathbf{A}^\top)$, so $\text{rank}(\mathbf{A}^\top \mathbf{A}) = \text{rank}(\mathbf{AA}^\top) = \text{rank}(\mathbf{A})$.
- (d) $\text{rank}(\mathbf{A} + \mathbf{C}) \leq \text{rank}(\mathbf{A}) + \text{rank}(\mathbf{C})$ for $\mathbf{C}$ of the same size as $\mathbf{A}$.

Proof.

- (a) is Eq. 1.1.2, together with $\text{rank}(\mathbf{AB}) = \text{rank}(\mathbf{B}^\top \mathbf{A}^\top) \leq \text{rank}(\mathbf{B}^\top)$ by the first inclusion and Theorem 1.2.1. (b) If $\mathbf{B}$ has full row rank, $\mathcal{C}(\mathbf{B}) = \mathbb{R}^n$ because it is an n -dimensional subspace of $\mathbb{R}^n$, so every $\mathbf{A}\mathbf{u}$ equals $\mathbf{AB}\mathbf{v}$ for some $\mathbf{v}$. If $\mathbf{A}$ has full column rank, $\mathbf{AB}\mathbf{v} = \mathbf{0}$ forces $\mathbf{B}\mathbf{v} = \mathbf{0}$. The rank statements follow from rank–nullity. (c) If $\mathbf{A}^\top \mathbf{A}\mathbf{v} = \mathbf{0}$ then $\|\mathbf{A}\mathbf{v}\|^2 = \mathbf{v}^\top \mathbf{A}^\top \mathbf{A}\mathbf{v} = 0$, so $\mathbf{A}\mathbf{v} = \mathbf{0}$. Thus the null spaces agree, and by Theorem 1.2.2 $\text{rank}(\mathbf{A}^\top \mathbf{A}) = \text{rank}(\mathbf{A})$. Since $\mathcal{C}(\mathbf{A}^\top \mathbf{A}) \subseteq \mathcal{C}(\mathbf{A}^\top)$ and both have dimension $\text{rank}(\mathbf{A})$, they are equal. Applying this to $\mathbf{A}^\top$ gives $\text{rank}(\mathbf{AA}^\top) = \text{rank}(\mathbf{A})$. (d) $\mathcal{C}(\mathbf{A} + \mathbf{C}) \subseteq \mathcal{C}(\mathbf{A}) + \mathcal{C}(\mathbf{C})$, and the dimension of a sum is at most the sum of the dimensions. ■

Part (c) is the reason the normal equations $\mathbf{X}^\top \mathbf{X}\mathbf{b} = \mathbf{X}^\top \mathbf{y}$ always have a solution: the right-hand side lies in $\mathcal{C}(\mathbf{X}^\top) = \mathcal{C}(\mathbf{X}^\top \mathbf{X})$ (Section 1.4).

1.2.4 Nonsingular submatrices

Proposition 1.2.4

Let $\mathbf{A}$ have rank $r \geq 1$. Choose any r linearly independent rows, indexed by I , and any r linearly independent columns, indexed by J . Then the $r \times r$ submatrix $\mathbf{A}_{IJ}$ at their intersection is nonsingular, and

$$\mathbf{A} = \mathbf{A}_{\cdot J} \mathbf{A}_{IJ}^{-1} \mathbf{A}_{I\cdot}, \quad (1.2.1)$$

where $\mathbf{A}_{\cdot J}$ holds the columns in J and $\mathbf{A}_{I\cdot}$ the rows in I .

Proof. The columns in J are a basis of $\mathcal{C}(\mathbf{A})$, so $\mathbf{A} = \mathbf{A}_{\cdot J} \mathbf{F}$ for some $r \times n$ matrix $\mathbf{F}$. Keeping only the rows in I gives $\mathbf{A}_{I\cdot} = \mathbf{A}_{IJ} \mathbf{F}$. The left side has rank r , so by Proposition 1.2.3(a) the $r \times r$ matrix $\mathbf{A}_{IJ}$ has rank r and is nonsingular. Then $\mathbf{F} = \mathbf{A}_{IJ}^{-1} \mathbf{A}_{I\cdot}$, which gives Eq. 1.2.1. ■

Conversely, a nonsingular $k \times k$ submatrix makes the k columns of $\mathbf{A}$ that contain it linearly independent, so $\text{rank}(\mathbf{A}) \geq k$. In particular, a matrix has rank r iff it has a nonsingular $r \times r$ submatrix and no nonsingular larger one. This proposition gives an explicit generalized inverse in Theorem 1.3.2.

1.2.5 Exercises

1.2.5.1 A. Check your understanding

Exercise A1.

Show that $\sum_{i < j} (y_i - y_j)^2 = n \sum_i (y_i - \bar{y})^2$, and write both sides as $\mathbf{y}^\top \mathbf{A} \mathbf{y}$ with $\mathbf{A}$ symmetric. Find the symmetric matrix of the quadratic form $2x_1^2 - x_2^2 + 6x_1x_2 - 4x_2x_3$.

Exercise A2.

Show that $\mathbf{X}^\top \mathbf{X} \mathbf{B} = \mathbf{X}^\top \mathbf{X} \mathbf{C}$ implies $\mathbf{X} \mathbf{B} = \mathbf{X} \mathbf{C}$. Deduce that $\mathbf{X}^\top \mathbf{X} \mathbf{b} = \mathbf{0}$ iff $\mathbf{X} \mathbf{b} = \mathbf{0}$.

Solution. Put $\mathbf{D} = \mathbf{B} - \mathbf{C}$, so that $\mathbf{X}^\top \mathbf{X} \mathbf{D} = \mathbf{0}$. Then $(\mathbf{X} \mathbf{D})^\top (\mathbf{X} \mathbf{D}) = \mathbf{D}^\top \mathbf{X}^\top \mathbf{X} \mathbf{D} = \mathbf{0}$, and by Eq. 1.1.3 a matrix $\mathbf{Z}$ with $\mathbf{Z}^\top \mathbf{Z} = \mathbf{0}$ is zero. So $\mathbf{X} \mathbf{D} = \mathbf{0}$. The second statement is the case $\mathbf{C} = \mathbf{0}$, with the converse obvious.

Exercise A3.

For the additive two-way layout with two rows, three columns and one observation per cell, write the 6×6 model matrix $\mathbf{X} = [\mathbf{1}, \mathbf{R}, \mathbf{C}]$, where $\mathbf{R}$ and $\mathbf{C}$ hold the row and

column indicators. Find $\text{rank}(\mathbf{X})$ and a basis of $\mathcal{N}(\mathbf{X})$. Compare $[\mathbf{R}, \mathbf{C}]^\top$ with the matrix $\mathbf{A}$ of Example (Filling a table from its margins).

1.2.5.2 B. Practice

Exercise B1.

Let $\mathbf{A}$ be $m \times n$ and $\mathbf{B}$ be $n \times q$. Prove $\text{rank}(\mathbf{AB}) \geq \text{rank}(\mathbf{A}) + \text{rank}(\mathbf{B}) - n$. *Hint:* apply rank-nullity to $\mathbf{AB}_0$, where the columns of $\mathbf{B}_0$ are a basis of $\mathcal{C}(\mathbf{B})$.

Solution. Let the $n \times s$ matrix $\mathbf{B}_0$, $s = \text{rank}(\mathbf{B})$, have columns forming a basis of $\mathcal{C}(\mathbf{B})$. Then $\mathcal{C}(\mathbf{AB}) = \mathcal{C}(\mathbf{AB}_0)$, so by rank-nullity $\text{rank}(\mathbf{AB}) = s - \dim \mathcal{N}(\mathbf{AB}_0)$. The map $\mathbf{c} \mapsto \mathbf{B}_0 \mathbf{c}$ is one-to-one and sends $\mathcal{N}(\mathbf{AB}_0)$ into $\mathcal{N}(\mathbf{A})$, so $\dim \mathcal{N}(\mathbf{AB}_0) \leq \dim \mathcal{N}(\mathbf{A}) = n - \text{rank}(\mathbf{A})$. Hence $\text{rank}(\mathbf{AB}) \geq \text{rank}(\mathbf{B}) - n + \text{rank}(\mathbf{A})$.

Exercise B2.

Show that if $\mathbf{A} = \mathbf{B}_1 \mathbf{F}_1 = \mathbf{B}_2 \mathbf{F}_2$ are two rank factorizations, then $\mathbf{B}_2 = \mathbf{B}_1 \mathbf{T}$ and $\mathbf{F}_2 = \mathbf{T}^{-1} \mathbf{F}_1$ for a nonsingular $\mathbf{T}$.